SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

SHORT ANSWER TEST

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1 - AT3

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DSA1520 - Cloud computing

Short Answer Test

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2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of data centers in delivering cloud services
5. Differentiate Iaas, Paas, Saas, and xaas with one suitable real-world example for each model.

ANSWERS

1. Cloud computing:

It is the delivery of computing services such as servers, storage, databases, networking and software through the internet instead of local computers

Four - main characteristics

(i) **On-demand self-service :-**

users can get computing resources whenever they need them without depending on a technician.

Example: creating a cloud storage account instantly

(ii) **Broad network access :**

Cloud services can be accessed through the internet using devices such as laptops, mobiles, etc.

(iii) **Resource pooling :**

Cloud providers share a large pool of resources such as servers and storage among many users.

(iv) **Rapid scalability**

Resources can be increased and decreased according to the demand.

Traditional premise	cloud computing
* Hardware is owned by the organization	* Resources are provided by the providers
* High initial cost	* Pay for resources used
* Manual scaling	* Quick and flexible scaling

2. Evolution of cloud computing.

1. Mainframe: Large mainframe computers were used by many users through terminals. This introduced the idea of sharing computer resources
2. Personal computers: computers became smaller & Cheaper. Users could perform computing skills and tasks.
3. Networking: computers were connected through LANs and WANs to share data and resources
4. Internet development: The internet connected the computers across the world. Services and information could be accessed remotly.
5. Virtualization: It allows one physical server to run multiple virtual machines. The improved hardware utilization and reduced cost
6. Web and internet software: Web applications became accessible through browsers. users did not need to install all software locally

7. Cloud computing: It is the combination of Internet connectivity, Virtualization, large data centers, automation and software technologies.

3. Server Virtualization:

* It is a technology that divides one physical server into multiple virtual servers or virtual machines (VMs).
* Each virtual machine can run its own operating system and applications.

Role in cloud computing.

- Improves the utilization of physical servers
- Allow multiple users to share the same physical hardware
- Reduce costs
- Easy to create and remove virtual machines.
- Scalability and flexibility.

Virtualization	cloud computing
• It is a technology	• It is a service delivery model
• creates virtual resources from physical resources	• provides computing resources to users
• Mainly focuses on resource utilization	• focuses on On-demand services
• can work without internet	• Usually accessed through internet.

A. Role of Data centers in delivering cloud services

* A data center is a facility containing a large number of servers, storage systems, networking devices, power system and cooling systems.
* Cloud providers use data center to deliver cloud services to users.

Role of Data centers

1. computing = provides servers to run applications and process data
2. Date storage = stores files, databases, applications and backups
3. Networking = connects cloud servers to users through high speed internet.
4. Security = protects data and systems using firewalls, authentications and monitoring
5. scalability = provides additional computing resources
6. Availability and Backups = Help to keep services available even when hardware fails.

5 Differentiation among Iaas, Paas, saas, xaas

Mode	full form	provides	Example.
Iaas	Infrastructure as a service	Virtual servers, storage and networking	Amazon EC2
Paas	Platform as a service	Platform and tools for developing applications	Google App Engine
Saas	Software as a service	Ready-to-use software	Google Docs
Xaas	Anything as a service	Almost any IT service through internet	AWS Lambda